

Answers

1. Normalize the following vector, $v = (3, 4)$

Answer: $|v| = \sqrt{3^2 + 4^2} = 5$
 $\hat{v} = (3/5, 4/5) = (0.6, 0.8)$

2. Normalize the following vector, $v = (-1, -1)$

Answer: $|v| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2} = 1.41\dots$
 $\hat{v} = (-1/1.41, -1/1.41) = (-0.707, -0.707)$

3. Normalize the following vector, $v = (0.5, 0.5)$

Answer: $|v| = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.5} = 0.707\dots$
 $\hat{v} = (0.5/0.707, 0.5/0.707) = (0.707, 0.707)$

4. Define a normalized or unit vector in your own words.

Answer: Generally, the idea here is to understand that a unit vector is simply a vector that's been scaled so that its length, or magnitude, is equal to 1. The direction remains the same.

5. Why is the length of a normalized vector always equal to 1?

Answer: This is the definition of a normalized vector, since it's obtained by dividing the original vector by the magnitude of the vector.

6. If a vector lies on the unit circle in a 2D grid, what can you infer about its magnitude?

Answer: The unit circle has a radius of 1, thus you can infer that the vector lies on the unit circle and is a unit vector, ie. has length 1.

7. If you have vectors $\vec{v}_1 = (0.75, 0.25)$ and $\vec{v}_2 = (3, 1)$, will the normalized vectors be the same?

Answer: Yes, $\vec{v}_2$ is simply $\vec{v}_1$ scaled by 4.

8. How do you denote a vector as unit length?

Answer: A unit vector is denoted with the "hat" symbol, like this: $\hat{v}$.